Enrolment No: 1025410s1386

ज्ञानेन प्रकाशते जगत्
INDUS UNIVERSITY

INDUS UNIVERSITY
Indus Institute of Technology & Engineering
Constituent Institute of Indus University

MID SEMESTER EXAMINATION (March 2026)

Institute:	IITE	**Exam Month / Year:**	March 2026
Program:	B. Tech.	**Branch:**	All
Semester:	IV	**Regular/Repeater:**	Regular
Subject Code:	ME0426	**Course Name:**	Robot's Mechanics of Motion (RMM) (OE)
Date:	24/03/26	**Time:**	90 Minutes
Day:	Tuesday	**Total Marks:**	40 Marks

Instructions:
1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

Q.1	**A**	Provide the scope and limitations of robots.	**04**
	B	Define the following with respect to robotics: i. Robot ii. Kinematics iii. Spherical robots iv. Kinematic Chains	**04**
		OR	
	B	What is visual perception? How does it work?	**04**
Q.2	**A**	Describe the steps involved in defining a point in the Cylindrical coordinate system.	**04**
	B	Differentiate Universal Frame and Body Frame with suitable examples.	**06**
		OR	
	B	Derive the homogeneous transformation matrix and explain its components.	**06**
	C	Explain the translation operator Trans(X, q) and state its properties.	**06**
		OR	
	C	Derive the Rotation Matrix and state its properties.	**06**
Q.3	**A**	State advantages and limitations of Euler angles in robotics.	**04**
	B	Briefly discuss Roll, Pitch and Yaw angles α, β and γ.	**06**
		OR	
	B	How the Denavit-Hartenberg approach is used in kinematic analysis of robots. Explain it in detail.	**06**
	C	Explain the difference between forward kinematics and inverse kinematics with a suitable example of a robotic arm.	**06**
		OR	
	C	Explain the concept of kinostatics in robotics. Illustrate how joint torques change when a robotic arm holds a load.	**06**

★★★

Scanned with OKEN Scanner